

Your Success is our win

Ex #2.1

(1)

Q#1) Specify the type of each of following matrices.

(i)
$$\begin{bmatrix} \sqrt{5} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & \frac{2}{5} \end{bmatrix}$$

Ans: Diagonal

(ii)
$$\begin{bmatrix} \sqrt{3} \\ 4 \\ 7 \end{bmatrix}$$

Ans: Column

(iii)
$$\begin{bmatrix} i/2 & 0 & i/3 \end{bmatrix}$$

Ans: Row

(iv)
$$\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$$

Ans: Symmetric

v)
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Ans: Identity / Symmetric

(vi)
$$\begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

Ans: Square / Symmetric

(vii)
$$\begin{bmatrix} \sqrt{5} & 0 & 0 \\ 0 & \sqrt{5} & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

Ans: Scalar / Symmetric

(viii)
$$\begin{bmatrix} 0 & i & 2i \\ -i & 0 & -4i \\ -2i & 4i & 0 \end{bmatrix}$$

Ans: Skew Hermitian

(ix)
$$\begin{bmatrix} 2 & -i & 5i \\ i & 3 & 7i \\ -5i & -7i & 4 \end{bmatrix}$$

Ans: Hermitian

Q#2) A newspaper agent of a town records the numbers of papers sold on each day of one week as follows.

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Daily Dawn	80	90	100	95	85	75	70
Daily Jung	100	110	90	95	105	85	80

Sol:
$$\begin{bmatrix} 80 & 90 & 100 & 95 & 85 & 75 & 70 \\ 100 & 110 & 90 & 95 & 105 & 85 & 80 \end{bmatrix}$$

2x7

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Ans: Identity / Symmetric

(vi)
$$\begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

Ans: Square / Symmetric

(vii)
$$\begin{bmatrix} \sqrt{7} & 0 & 0 \\ 0 & \sqrt{7} & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

Ans: Scalar / Symmetric

(viii)
$$\begin{bmatrix} 0 & i & 2i \\ -i & 0 & -4i \\ -2i & 4i & 0 \end{bmatrix}$$

Ans: Skew Hermitian

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$$\begin{bmatrix} 80 & 90 & 100 & 95 & 85 & 75 & 70 \\ 100 & 110 & 90 & 95 & 105 & 85 & 80 \end{bmatrix}$$

2x7

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(2)

Q#3) Find values of unknown in each of following-

$$(i) \begin{bmatrix} a & -4i \\ 5i & 6i \end{bmatrix} = \begin{bmatrix} 7i & b \\ c & d \end{bmatrix} \quad (ii) \begin{bmatrix} 2 & -3 & 5 \\ a & 9 & 0 \\ b & c & -1 \end{bmatrix} = \begin{bmatrix} d & e & g \\ -2 & f & u \\ -4 & 7 & -1 \end{bmatrix}$$

Sol

$$\Rightarrow a = 7i$$

$$\Rightarrow b = -4i$$

$$\Rightarrow c = 8i$$

$$\Rightarrow d = 6i$$

Sol

$$\Rightarrow a = -2$$

$$\Rightarrow b = -4$$

$$\Rightarrow c = 7$$

$$\Rightarrow d = 3$$

$$\Rightarrow e = -3$$

$$\Rightarrow f = 5$$

$$\Rightarrow g = 9$$

$$(iii) \begin{bmatrix} x+y & 0 & z \\ 9 & 2x+y & 6 \\ a & 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 3x \\ 9 & 4 & 6 \\ \frac{y}{2} & 1 & 3 \end{bmatrix}$$

Sol

$$\Rightarrow y = 3 - 1$$

$$\Rightarrow y = 2$$

$$\Rightarrow x + y = 3 \text{ --- (1)}$$

$$\Rightarrow 2x + y = 4 \text{ --- (2)}$$

$$\Rightarrow z = 3x \text{ --- (3)}$$

$$\Rightarrow a = \frac{y}{2} \text{ --- (4)}$$

$$\text{Put } x = 1 \text{ in (3)}$$

$$\Rightarrow z = 3(1)$$

$$\Rightarrow z = 3$$

Sub (2) from (1)

$$x + y = 3$$

$$-2x + y = 4$$

$$+x = -4$$

$$\Rightarrow x = -4$$

$$\text{Put } y = 2 \text{ in (4)}$$

$$\Rightarrow a = \frac{2}{2}$$

$$\Rightarrow a = 1$$

$$\text{Put } x = 1 \text{ in (1)}$$

$$\Rightarrow 1 + y = 3$$

Thus,

$$x = 1, y = 2, z = 3, a = 1$$

(3)

Q#4) Find the transpose of each of the following matrices

$$(i) \begin{bmatrix} 4 & 3 & 6 \end{bmatrix}$$

Sol

$$A = \begin{bmatrix} 4 & 3 & 6 \end{bmatrix}_{1 \times 3}$$

$$A^T = \begin{bmatrix} 4 \\ 3 \\ 6 \end{bmatrix}_{3 \times 1}$$

$$(iv) \begin{bmatrix} 2i & 7 \\ 5i & 7i \\ 2i & -5i \end{bmatrix}$$

Sol

$$A = \begin{bmatrix} 2i & 7 \\ 5i & 7i \\ 2i & -5i \end{bmatrix}_{3 \times 2}$$

$$A^T = \begin{bmatrix} 2i & 5i & 2i \\ 7 & 7i & -5i \end{bmatrix}_{2 \times 3}$$

$$(ii) \begin{bmatrix} 2i & 5i & -3i \\ 0 & -6i & 2i \end{bmatrix}$$

Sol

$$A = \begin{bmatrix} 2i & 5i & -3i \\ 0 & -6i & 2i \end{bmatrix}_{2 \times 3}$$

$$A^T = \begin{bmatrix} 2i & 0 \\ 5i & -6i \\ -3i & 2i \end{bmatrix}_{3 \times 2}$$

$$(v) \begin{bmatrix} -8 & 7 \\ 3 & 0 \end{bmatrix}$$

Sol

$$A = \begin{bmatrix} -8 & 7 \\ 3 & 0 \end{bmatrix}_{2 \times 2}$$

$$A^T = \begin{bmatrix} -8 & 3 \\ 7 & 0 \end{bmatrix}_{2 \times 2}$$

$$(iii) \begin{bmatrix} 8 \\ 3 \\ -4 \end{bmatrix}$$

Sol

$$A = \begin{bmatrix} 8 \\ 3 \\ -4 \end{bmatrix}_{3 \times 1}$$

$$A^T = \begin{bmatrix} 8 & 3 & -4 \end{bmatrix}_{1 \times 3}$$

$$(vi) \begin{bmatrix} -7 & -10 & 8 \\ 4 & 5 & 9 \\ -1 & 2 & 3 \end{bmatrix}$$

Sol

$$A = \begin{bmatrix} -7 & -10 & 8 \\ 4 & 5 & 9 \\ -1 & 2 & 3 \end{bmatrix}_{3 \times 3}$$

$$A^T = \begin{bmatrix} -7 & 4 & -1 \\ -10 & 5 & 2 \\ 8 & 9 & 3 \end{bmatrix}_{3 \times 3}$$

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(2)

Q#3) Find values of unknown in each of following-

$$(i) \begin{bmatrix} a & -4i \\ 5i & 6i \end{bmatrix} = \begin{bmatrix} 7i & b \\ c & d \end{bmatrix} \quad (ii) \begin{bmatrix} 2 & -3 & 5 \\ a & 9 & 0 \\ b & c & -1 \end{bmatrix} = \begin{bmatrix} d & e & g \\ -2 & f & u \\ -4 & 7 & -1 \end{bmatrix}$$

Sol

$$\Rightarrow a = 7i$$

$$\Rightarrow b = -4i$$

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Sol

$$\Rightarrow a = -2$$

$$\Rightarrow b = -4$$

$$\Rightarrow c = 7$$

$$\Rightarrow d = 3$$

$$\Rightarrow e = -3$$

$$\Rightarrow g = 5$$

$$\Rightarrow f = 9$$

$$(iii) \begin{bmatrix} x+y & 0 & z \\ 9 & 2x+y & 6 \\ a & 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 3x \\ 9 & 4 & 6 \\ \frac{y}{2} & 1 & 3 \end{bmatrix}$$

Sol

$$\Rightarrow x+y = 3 \text{ --- (1)}$$

$$\Rightarrow 2x+y = 4 \text{ --- (2)}$$

$$\Rightarrow z = 3x \text{ --- (3)}$$

$$\Rightarrow a = \frac{y}{2} \text{ --- (4)}$$

$$\Rightarrow y = 3 - 1$$

$$\Rightarrow y = 2$$

$$\text{Put } x = 1 \text{ in (3)}$$

$$\Rightarrow z = 3(1)$$

$$\Rightarrow z = 3$$

Sub (2) from (1)

$$x + \frac{y}{2} = 3$$

$$2x + y = 4$$

$$+x = +4$$

$$\Rightarrow x = 1$$

$$\text{Put } y = 2 \text{ in (4)}$$

$$\Rightarrow a = \frac{2}{2}$$

$$\Rightarrow a = 1$$

Thus,

$$x = 1, y = 2, z = 3, a = 1$$

$$\text{Put } x = 1 \text{ in (1)}$$

$$\Rightarrow 1 + y = 3$$

(3)

Q#4) Find the transpose of each of the following matrices

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Sol

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$$(iv) \begin{bmatrix} 2i & 7 \\ 5i & 7i \\ 2i & -5i \end{bmatrix}$$

Sol

$$A = \begin{bmatrix} 2i & 7 \\ 5i & 7i \\ 2i & -5i \end{bmatrix}_{3 \times 2}$$

$$A^T = \begin{bmatrix} 2i & 5i & 2i \\ 7 & 7i & -5i \end{bmatrix}_{2 \times 3}$$

$$(ii) \begin{bmatrix} 2i & 5i & -3i \\ 0 & -6i & 2i \end{bmatrix}$$

Sol

$$A = \begin{bmatrix} 2i & 5i & -3i \\ 0 & -6i & 2i \end{bmatrix}_{2 \times 3}$$

$$A^T = \begin{bmatrix} 2i & 0 \\ 5i & -6i \\ -3i & 2i \end{bmatrix}_{3 \times 2}$$

$$(v) \begin{bmatrix} -8 & 7 \\ 3 & 0 \end{bmatrix}$$

Sol

$$A = \begin{bmatrix} -8 & 7 \\ 3 & 0 \end{bmatrix}_{2 \times 2}$$

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$$A = \begin{bmatrix} 8 \\ 3 \\ -4 \end{bmatrix}_{3 \times 1}$$

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$$A = \begin{bmatrix} -7 & -10 & 8 \\ 4 & 5 & 9 \\ -1 & 2 & 3 \end{bmatrix}_{3 \times 3}$$

$$A^T = \begin{bmatrix} -7 & 4 & -1 \\ -10 & 5 & 2 \\ 8 & 9 & 3 \end{bmatrix}_{3 \times 3}$$

⑤ Write down in tabular form:

(i) $A = [a_{ij}]_{(2,3)}$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

(ii) $X = [x_{ij}]_{(3,4)}$

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \end{bmatrix}$$

(iii) $B = [b_{ij}]_{(4,4)}$

$$B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \\ b_{41} & b_{42} & b_{43} & b_{44} \end{bmatrix}$$

⑥ Which of the following are symmetric or skew-symmetric matrices.

(i) $\begin{bmatrix} 0 & -5 & -6 \\ 5 & 0 & 7 \\ 6 & -7 & 0 \end{bmatrix}$

$$A = \begin{bmatrix} 0 & -5 & -6 \\ 5 & 0 & 7 \\ 6 & -7 & 0 \end{bmatrix}$$

$$A^t = \begin{bmatrix} 0 & 5 & 6 \\ -5 & 0 & -7 \\ -6 & 7 & 0 \end{bmatrix}$$

Given matrix is Anti-symmetric.

(ii) $\begin{bmatrix} 7 & 5 & 8 \\ 5 & -1 & 6 \\ 8 & 6 & -1 \end{bmatrix}$

$$A = \begin{bmatrix} 7 & 5 & 8 \\ 5 & -1 & 6 \\ 8 & 6 & -1 \end{bmatrix}$$

$$A^t = \begin{bmatrix} 7 & 5 & 8 \\ 5 & -1 & 6 \\ 8 & 6 & -1 \end{bmatrix}$$

Given matrix is Symmetric.

(iii) $\begin{bmatrix} 0 & 1 & 3 \\ -1 & 0 & 5 \\ -3 & 5 & 0 \end{bmatrix}$

$$A = \begin{bmatrix} 0 & 1 & 3 \\ -1 & 0 & 5 \\ -3 & 5 & 0 \end{bmatrix}$$

$$A^t = \begin{bmatrix} 0 & -1 & -3 \\ 1 & 0 & -5 \\ 3 & 5 & 0 \end{bmatrix}$$

Given matrix is anti-symmetric.

(iv) $\begin{bmatrix} 0 & -5 & 4 \\ 5 & 0 & -1 \\ -4 & 1 & 0 \end{bmatrix}$

$$A = \begin{bmatrix} 0 & -5 & 4 \\ 5 & 0 & -1 \\ -4 & 1 & 0 \end{bmatrix}$$

$$A^t = \begin{bmatrix} 0 & 5 & -4 \\ -5 & 0 & 1 \\ 4 & -1 & 0 \end{bmatrix}$$

Given matrix is sq anti-symmetric.

(v) $\begin{bmatrix} 1 & 2 & 5 \\ 2 & 5 & -7 \\ 5 & -7 & 3 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 5 & -7 \\ 5 & -7 & 3 \end{bmatrix}$$

$$A^t = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 5 & -7 \\ 5 & -7 & 3 \end{bmatrix}$$

Given matrix is symmetric.

⑥

Q4) Find index of nilpotent matrices.

a)
$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Sol

let, $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$

$A^2 = A \cdot A$

$A^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$

$A^2 = \begin{bmatrix} (0)(0) + (1)(0) + (0)(0) & (0)(1) + (1)(0) + (0)(0) & (0)(0) + (1)(1) + (0)(0) \\ (0)(0) + (0)(0) + (1)(0) & (0)(1) + (0)(0) + (1)(0) & (0)(0) + (0)(1) + (1)(0) \\ (0)(0) + (0)(0) + (0)(0) & (0)(1) + (0)(0) + (0)(0) & (0)(0) + (0)(1) + (0)(0) \end{bmatrix}$

$A^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$

$A^3 = A^2 \cdot A$

$A^3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$

$A^3 = \begin{bmatrix} (0)(0) + (0)(0) + (1)(0) & (0)(1) + (0)(0) + (1)(0) & (0)(0) + (0)(1) + (1)(0) \\ (0)(0) + (0)(0) + (0)(0) & (0)(1) + (0)(0) + (0)(0) & (0)(0) + (0)(1) + (0)(0) \\ (0)(0) + (0)(0) + (0)(0) & (0)(1) + (0)(0) + (0)(0) & (0)(0) + (0)(1) + (0)(0) \end{bmatrix}$

$A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$

$A^3 = 0$, so A is a nilpotent matrix of index 3.

⑦

(ii)
$$\begin{bmatrix} 1 & -3 & -4 \\ 1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$$

Sol

let $A = \begin{bmatrix} 1 & -3 & -4 \\ 1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}_{3 \times 3}$

$A^2 = A \cdot A$

$A^2 = \begin{bmatrix} 1 & -3 & -4 \\ 1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 & -3 & -4 \\ 1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}_{3 \times 3}$

$A^2 = \begin{bmatrix} (1)(1) + (-3)(1) + (-4)(1) & (1)(-3) + (-3)(3) + (-4)(-3) & (1)(-4) + (-3)(4) + (-4)(-4) \\ (-1)(1) + (3)(1) + (4)(1) & (-1)(-3) + (3)(3) + (4)(-3) & (-1)(-4) + (3)(4) + (4)(-4) \\ (1)(1) + (-3)(1) + (-4)(1) & (1)(-3) + (-3)(3) + (4)(-3) & (1)(-4) + (-3)(4) + (-4)(-4) \end{bmatrix}$

$A^2 = \begin{bmatrix} 1+3-4 & -3-9+12 & -4-12+16 \\ -1+3+4 & 3+9-12 & 4+12-16 \\ 1+3-4 & -3-9+12 & -4-12+16 \end{bmatrix}_{3 \times 3}$

$A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$

$A^2 = 0$, so A is nilpotent matrix of index 2.

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(iii)

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$$

Sol

$$\text{let, } A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}_{3 \times 3}$$

$$A^2 = A \cdot A$$

$$A^2 = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}_{3 \times 3}$$

$$A^2 = \begin{bmatrix} (1)(1)+(1)(5)+(3)(-2) & (1)(1)+(1)(2)+(3)(-1) & (1)(3)+(1)(6)+(3)(-3) \\ (5)(1)+(2)(5)+(6)(-2) & (5)(1)+(2)(2)+(6)(-1) & (5)(3)+(2)(6)+(6)(-3) \\ (-2)(1)+(-1)(5)+(-3)(-2) & (-2)(1)+(-1)(2)+(-3)(-1) & (-2)(3)+(-1)(6)+(-3)(-3) \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1+5-6 & 1+2-3 & 3+6-9 \\ 5+10-12 & 5+4-6 & 15+12-18 \\ -2-5+6 & -2-2+3 & -6-6+9 \end{bmatrix}_{3 \times 3}$$

$$A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 3 \\ -1 & -1 & -3 \end{bmatrix}_{3 \times 3}$$

$$A^3 = A^2 \cdot A$$

$$A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 3 \\ -1 & -1 & -3 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}_{3 \times 3}$$

$$A^3 = \begin{bmatrix} (0)(1)+(0)(5)+(0)(-2) & (0)(1)+(0)(2)+(0)(-1) & (0)(3)+(0)(6)+(0)(-3) \\ (3)(1)+(3)(5)+(3)(-2) & (3)(1)+(3)(2)+(3)(-1) & (3)(3)+(3)(6)+(3)(-3) \\ (-1)(1)+(-1)(5)+(-3)(-2) & (-1)(1)+(-1)(2)+(-3)(-1) & (-1)(3)+(-1)(6)+(-3)(-3) \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 0+0+0 & 0+0+0 & 0+0+0 \\ 3+15-12 & 3+6-9 & 9+18-27 \\ -1-5+6 & -1-2+3 & -3-6+9 \end{bmatrix}$$

Sol

$$A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$$

$A^3 = 0$, So A is a nilpotent matrix of index 3.

③ Find the period of following periodic matrix.

$$\textcircled{1} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$$

Sol

$$\text{let, } A = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}_{3 \times 3}$$

$$\therefore A^{k+1} = A$$

$$\Rightarrow k = 1, \text{ we have } A^2$$

$$A^2 = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}_{3 \times 3}$$

$$A^2 = \begin{bmatrix} (1)(1)+(-2)(-3)+(-6)(2) & (1)(-2)+(-2)(2)+(-6)(0) & (1)(-6)+(-2)(9)+(-6)(-3) \\ (-3)(1)+(2)(-3)+(9)(2) & (-3)(-2)+(2)(2)+(9)(0) & (-3)(-6)+(2)(9)+(9)(-3) \\ (2)(1)+(0)(-3)+(-3)(2) & (2)(-2)+(0)(2)+(-3)(0) & (2)(-6)+(0)(9)+(-3)(-3) \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1+6-12 & -2-4-0 & -6-18+18 \\ -3-6+18 & 6+4+0 & 18+18-27 \\ 2+0-6 & -4+0+0 & -12+0+9 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix}_{3 \times 3}$$

9

(10)

For $K=2$, we have A^2 .

$$\Rightarrow A^2 = A \cdot A$$

$$\Rightarrow A^2 = \begin{bmatrix} -5 & -7 & 0 \\ 3 & 5 & 0 \\ -4 & -4 & -3 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} (-5)(1) + (-7)(-3) + (0)(2) & (-5)(-2) + (-7)(2) + (0)(0) & (-5)(-6) + (-7)(9) + (0)(-3) \\ (3)(1) + (5)(-3) + (0)(2) & (3)(-2) + (5)(2) + (0)(0) & (3)(-6) + (5)(9) + (0)(-3) \\ (-4)(1) + (-4)(-3) + (-3)(2) & (-4)(-2) + (-4)(2) + (-3)(0) & (-4)(-6) + (-4)(9) + (-3)(-3) \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} -5 + 21 - 0 & 10 - 14 + 0 & 30 - 63 + 0 \\ 3 - 15 + 0 & -6 + 10 + 0 & -18 + 45 + 0 \\ -4 + 12 - 6 & 8 - 8 + 0 & 24 - 36 + 9 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}_{3 \times 3}$$

$A^2 = A$, So A is a periodic matrix of period 2.

Q) Which of the following is idempotent or involutory.

$$(i) \begin{bmatrix} -5 & -7 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

$$\text{let, } A = \begin{bmatrix} -5 & -7 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}_{3 \times 3}$$

(11)

$$\Rightarrow A^2 = A \cdot A$$

$$\Rightarrow A^2 = \begin{bmatrix} -5 & -7 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}_{3 \times 3} \begin{bmatrix} -5 & -7 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} (-5)(-5) + (-7)(3) + (0)(1) & (-5)(-7) + (-7)(5) + (0)(2) & (-5)(0) + (-7)(0) + (0)(-1) \\ (3)(-5) + (5)(3) + (0)(1) & (3)(-7) + (5)(5) + (0)(2) & (3)(0) + (5)(0) + (0)(-1) \\ (1)(-5) + (2)(3) + (-1)(1) & (1)(-7) + (2)(5) + (-1)(2) & (1)(0) + (2)(0) + (-1)(-1) \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 25 - 21 + 0 & 21 - 35 + 0 & 0 + 0 + 0 \\ -15 + 15 + 0 & -21 + 25 + 0 & 0 + 0 + 0 \\ -5 + 6 - 1 & -7 + 10 - 2 & 0 + 0 + 1 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} 4 & -14 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

As $A^2 \neq I$, so A is an involutory matrix.

$$(ii) \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

Sol

$$\text{let, } A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = A \cdot A$$

$$\Rightarrow A^2 = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} (2)(2) + (-2)(-1) + (-4)(1) & (2)(-2) + (-2)(3) + (-4)(-2) & (2)(-4) + (-2)(4) + (-4)(-3) \\ (-1)(2) + (3)(-1) + (4)(1) & (-1)(-2) + (3)(3) + (4)(-2) & (-1)(-4) + (3)(4) + (4)(-3) \\ (1)(2) + (-2)(-1) + (-3)(1) & (1)(-2) + (-2)(3) + (-3)(-2) & (1)(-4) + (-2)(4) + (-3)(-3) \end{bmatrix}$$

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$$\Rightarrow A^2 = \begin{bmatrix} 4+2-4 & -4-6+8 & -9-8+12 \\ -2-3+4 & 2+9-8 & 4+12-12 \\ 2+2-3 & -2-6+6 & -4-8+9 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} 2 & -2 & 12 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}_{3 \times 3}$$

As $A^2 = A$, so A is a idempotent matrix.

(11) Which of the following is Hermitian or skew-Hermitian Matrix.

(i) $\begin{bmatrix} 3 & 1-2i & 4+7i \\ 1+2i & -4 & -2i \\ 4-7i & 2i & 5 \end{bmatrix}$

(ii) $\begin{bmatrix} -i & 3i & i \\ 3i & 7i & -5i \\ i & -5i & -i \end{bmatrix}$

let $A = \begin{bmatrix} 3 & 1-2i & 4+7i \\ 1+2i & -4 & -2i \\ 4-7i & 2i & 5 \end{bmatrix}_{3 \times 3}$

First we find $\bar{A}$:

$$\bar{A} = \begin{bmatrix} 3 & 1+2i & 4-7i \\ 1-2i & -4 & 2i \\ 4+7i & -2i & 5 \end{bmatrix}_{3 \times 3}$$

Now,

$$(\bar{A})^t = \begin{bmatrix} 3 & 1+2i & 4-7i \\ 1-2i & -4 & 2i \\ 4+7i & -2i & 5 \end{bmatrix}_{3 \times 3}$$

As $(\bar{A})^t = A$, so A is a Hermitian Matrix.

(ii)

let, $A = \begin{bmatrix} -i & 3i & i \\ 3i & 7i & -5i \\ i & -5i & -i \end{bmatrix}_{3 \times 3}$

First we find $\bar{A}$:

$$\bar{A} = \begin{bmatrix} i & -3i & -i \\ -3i & -7i & 5i \\ -i & 5i & i \end{bmatrix}_{3 \times 3}$$

Now,

$$(\bar{A})^t = \begin{bmatrix} i & -3i & -i \\ -3i & -7i & 5i \\ -i & 5i & i \end{bmatrix}_{3 \times 3}$$

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(i) wht $(\bar{A})^t = - \begin{bmatrix} -i & 3i & i \\ 3i & 7i & -5i \\ i & -5i & -i \end{bmatrix}_{3 \times 3}$

As $(\bar{A})^t = A$, so A is skew Hermitian matrix.

(iii) $\begin{bmatrix} 1 & -i & -1+i \\ i & 1 & 1+i \\ 1+i & -1+i & 0 \end{bmatrix}$

(i)

let, $A = \begin{bmatrix} 1 & -i & -1+i \\ i & 1 & 1+i \\ 1+i & -1+i & 0 \end{bmatrix}_{3 \times 3}$

First we find $\bar{A}$:

$$\bar{A} = \begin{bmatrix} 1 & i & -1-i \\ -i & 1 & 1-i \\ 1-i & -1-i & 0 \end{bmatrix}_{3 \times 3}$$

$$(\bar{A})^t = \begin{bmatrix} 1 & -i & 1-i \\ i & 1 & -1-i \\ -1-i & 1-i & 0 \end{bmatrix}_{3 \times 3}$$

As $(\bar{A})^t \neq A$, so A is neither Hermitian matrix nor skew Hermitian Matrix.

(11) Find real numbers x, y, z such that matrix A is Hermitian Matrix.

$$A = \begin{bmatrix} 3 & x+2i & yi \\ 3-2i & 0 & 1+2i \\ yi & 1-xi & -1 \end{bmatrix}$$

(i)

(14)

$$\bar{A} = \begin{bmatrix} 3 & x-2i & -yi \\ 3+2i & 0 & 1-2i \\ -4i & 1+xi & -1 \end{bmatrix} \quad 3 \times 3$$

$$(\bar{A})^t = \begin{bmatrix} 3 & 3+2i & -4i \\ x-2i & 0 & 1-2i \\ -yi & 1+xi & -1 \end{bmatrix} \quad 3 \times 3$$

For Hermitian Matrix, $A = (\bar{A})^t$

$$\Rightarrow A = (\bar{A})^t$$

$$\Rightarrow \begin{bmatrix} 3 & x+2i & yi \\ 3-2i & 0 & 1+2i \\ 4i & 1-xi & -1 \end{bmatrix} = \begin{bmatrix} 3 & 3+2i & -4i \\ x-2i & 0 & 1+xi \\ -4i & 1-2i & -1 \end{bmatrix}$$

$$x+2i = 3+2i \quad \text{--- (1)}$$

$$1+2i = 1+xi \quad \text{--- (4)}$$

$$yi = -4i \quad \text{--- (2)}$$

$$1-xi = 1-2i \quad \text{--- (5)}$$

$$3-2i = x-2i \quad \text{--- (3)}$$

$$yi = -4i \quad \text{--- (6)}$$

From (1)

$$x+2i = 3+2i$$

$$x = 3$$

From (4)

$$1+2i = 1+xi$$

$$2i = xi$$

From (2)

$$2i = 3i$$

$$yi = -4i$$

$$2 = 3$$

$$yi + 4i = 0$$

$$y(2i) = 0$$

$$y = 0$$

$$2 = 3$$

Your Success is our Win (15)

From (5)

$$x-xi = x-2i$$

$$13x = x2x$$

$$2 = 3$$

From (3)

$$2-yi = x-2i$$

$$x = 2$$

Thus, $x = 3$, $y = 0$, $2 = 3$ E.P.

Ex # 2.2

$$\text{If } A = \begin{bmatrix} 1+i & 2 & 3i \\ 4i & 5-i & 2+3i \\ 0 & 5 & 1-i \end{bmatrix} \text{ and } B = \begin{bmatrix} 2+i & 3-i & 4 \\ i & 0 & 5-i \\ 6+i & 2 & 2+3i \end{bmatrix}$$

Sol

(i) $A+B$

$$\Rightarrow A+B = \begin{bmatrix} 1+i & 2 & 3i \\ 4i & 5-i & 2+3i \\ 0 & 5 & 1-i \end{bmatrix} + \begin{bmatrix} 2+i & 3-i & 4 \\ i & 0 & 5-i \\ 6+i & 2 & 2+3i \end{bmatrix}$$

3x3

3x3

$$\Rightarrow A+B = \begin{bmatrix} 1+i+2+i & 2+3-i & 3i+4 \\ 4i+i & 5-i+0 & 2+3i+5-i \\ 0+6+i & 5+2 & 1-i+2+3i \end{bmatrix}$$

3x3

$$\Rightarrow A \times B = \begin{bmatrix} 3+2i & 5-i & 3i+4 \\ 5i & 5-i & 7+2i \\ 6+i & 7 & 3+3i \end{bmatrix}$$

3x3